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**5th Sem / Mech, Prod, Mecatronics (4th sem),
CAD/CAM, Mech. Engg. (Fabrication Tech.)
Mechanical Engg. (Cad/CAM Design & Robotics)
Sub : Theory of Machines**

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 In higher pairs, there is :

- a) Surface contact b) Line or point contact
- c) No contact d) Rigid contact

Q.2 Velocity ratio in belt drive is affected by :

- a) Slip b) Module
- c) Pitch circle d) Addendum

Q.3 The condition for maximum power transmission in belt drive is :

- a) $T_2 = 0$
- b) Centrifugal tension = $\frac{1}{3}$ max tension
- c) $T_1 = T_2$
- d) $\mu = 0$

Q.4 Coefficient of fluctuation of speed is :

- a) $(N_1 - N_2)/N$ b) $(N_1 - N_2)/N$
- c) N_1/N_2 d) $N_2 - N_1$

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- Q.5 Watt governor is suitable for :
- a) High speed b) Low speed
 - c) Constant speed d) Variable load
- Q.6 Base circle in cam is :
- a) Smallest circle of cam
 - b) Largest circle
 - c) Pitch curve
 - d) Prime circle
- Q.7 Dynamic balancing requires:
- a) Resultant force = 0
 - b) Resultant couple = 0
 - c) Both Zero
 - d) None of the above
- Q.8 Harmful effects of vibration include:
- a) Noise b) Wear
 - c) Fatigue failure d) All of these
- Q.9 Oldham coupling is inversion of :
- a) Single slider crank chain
 - b) Double slider crank chain
 - c) Four bar chain
 - d) Cam mechanism
- Q.10 A followers may be :
- a) Knife edge b) Roller
 - c) Flat Faced d) All of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define kinematic chain.
- Q.12 Generally belts are made from which material?
- Q.13 Hartnell governor is a _____ type of governor.
- Q.14 Define crowning of pulley.
- Q.15 Define Coefficient of fluctuation of speed.
- Q.16 What is Module?
- Q.17 What is dynamic equilibrium?
- Q.18 Define epicyclic gear train.
- Q.19 Define vibration.
- Q.20 Define cam.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain types of kinematic pairs with examples.
- Q.22 What are the advantages of V-Belt over the flat belt pulley drive?
- Q.23 Differentiate between flywheel and governor.
- Q.24 Explain balancing in different planes.
- Q.25 Discuss harmful effects of vibrations.
- Q.26 Explain turning moment diagram for I.C. engine.

- Q.27 Explain types of followers.
- Q.28 Derive expression for tension ratio in flat belt.
- Q.29 Explain working of Porter governor.
- Q.30 Compare simple and compound gear train.
- Q.31 Explain slip and creep in belt drive.
- Q.32 Explain cam terminology.
- Q.33 Explain classification of gears.
- Q.34 Explain working of Oldham's coupling.
- Q.35 Define vibration & its types.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain Double slider crank chain. Also explain any two inversions of Double Slider Crank chain.
- Q.37 Explain the working and construction of Hartnell Governor with the help of neat sketch.
- Q.38 For a V-belt drive, derive the expression for tensions ratio of tight side and slack side.